Lab1 COMSM0123 Network Security
Message Analysis using Wireshark

In this lab, you will learn capture messages (Application, TCP segments, IP Packets and Ethernet Frames) using Wireshark. Also, you will be able to build, define and exercise Wireshark Message/Packet Display Filter (Typing Text, Expression Builder & Shortcut options). You will also compare the standards of the TCP/IP Layered Network Architecture with Wireshark captured HTTP and TCP messages.

Please read the **instructions** before solving any questions.

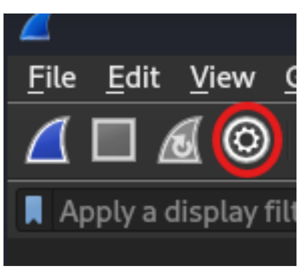
1. You can use your laptop (Linux system) (it is expected that you use Ubuntu for desktops: http://www.ubuntu.com/ .) And VMware which is open source on Linux systems.
2. You will be working individually on this lab
3. **Deliverable**: Report (Answers to the questions)

- **Notes:**
  - Suspend all unused VMs to maintain a decent performance on the desktop. The more active VMs, the more demands on the CPU and RAM.

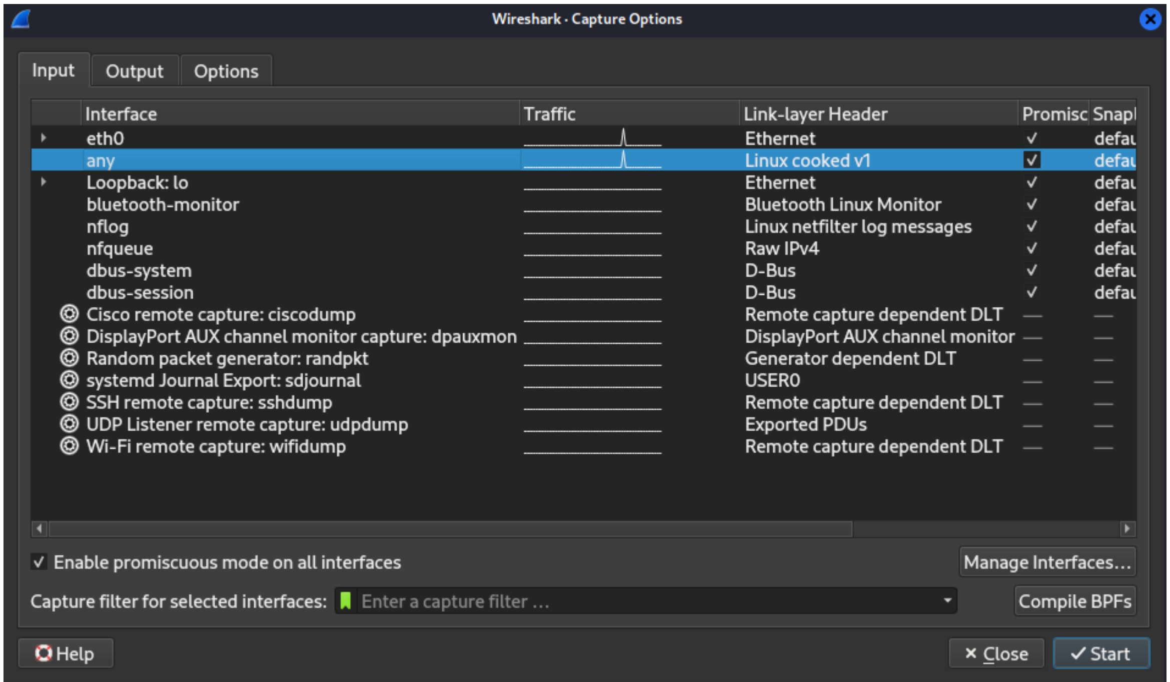
1. Exercise 1: Wireshark Setup/Capture (Host)

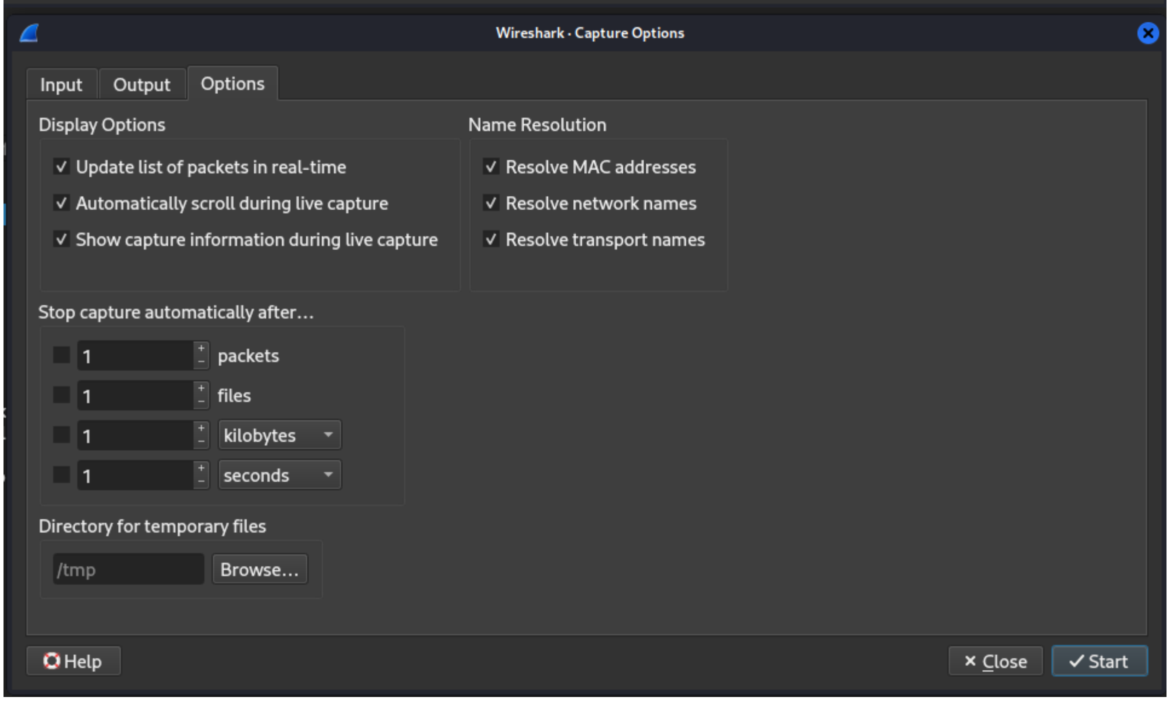
Make sure that the Host Hardware/External Network Interface ethx (where x is an integer) is connected to the public network (Internet), that is, it acquired an IP address.

Launch Wireshark on the Host by typing **sudo wireshark** in a terminal on the Host. Click on the Interfaces option in the Capture menu. Click on the Options button corresponding to the IP address acquired by the Host Hardware/External Network Interface ethx.



Check the boxes for all Displays Options and Name Resolution. Make sure that 'Capture all in promiscuous mode' is checked in. Click 'Start'.





□ Q1 - What does promiscuous mode mean?

Wireshark should start displaying ‘packets’ (which are actually ‘frames’) transmitted or received on the selected interface. Note that each line represents an ethernet frame. The Wireshark window should be divided into 3 panes, if not, you may have to click on a horizontal divider to show hidden panes.
The top pane displays one row of info for each frame/packet captured.

□ Q2 - Describe the information provided and explain the headings of the columns.

Note that the frames/packets (rows) are sorted by Time. However, you can change that by clicking on the heading of another column and therefore sort by the heading of that column.

□ Q3 – Provide an example where you have sorted the frames using the Protocol column.

The centre pane displays the protocols associated with the selected frame and the bottom pane displays the hexadecimal representation (Hex View) of the data contained in the same. Beneath the bottom pane is a status bar that displays the total number of Frames captured, displayed and marked (zero unless filtering is applied).

For example, if you select in the centre pane the Ethernet II layer (Data Link) of a frame, the hexadecimal representation of that layer information should be highlighted in the bottom pane. Take a screenshot.

□ Q4 – Find out and describe how one could change the bottom pane view from Hex View to Bits.

□ Q5 – Explain the how one would go about converting decimal number 87 into its Hex and Binary representations.

Exercise 2: Capturing Frames and encoding

Open a Firefox browser on the Host. Select Edit > Preferences > Privacy and select 'never remember history' and click on 'clear' all current history. Click 'close'. Make sure that the startup home page of your browser is Google or Ubuntu Google. Close the browser.

On Wireshark application (assuming it is still running!), select Capture > Restart (the yellow arrow) from the top menu bar. All captured and displayed Frames should be cleared from the top pane. Note – Almost instantaneously new frames are captured, the network is busy all the time, auto discover, etc.

In a terminal (on the Host) type **nslookup google.com** to force the generation of Domain Name requests.

□ Q6 – compare the results of nslookup on the Host terminal and the content of the corresponding DNS response message as captured by Wireshark.

Then launch Firefox browser on the Host to force the generation of http requests & responses and TCP segments. Note that ARP messages are dynamically generated regularly. Wireshark should display frames/packets with protocols, such as, ARP, DNS, TCP, HTTP, etc. Wait a second or two and then close the browser. Stop the capture by clicking on the stop icon (see above) or select Capture > Stop from the top menu bar. Save the captured packets in a file: select File > Save As > enter a name like: **Lab3_Capture_Lastnames.pcap**, save this file somewhere easy to access, like 'documents'.

□ Q7 – provide a copy of Lab3_Capture_Lastnames.pcap as part of your Lab 3

deliverables.

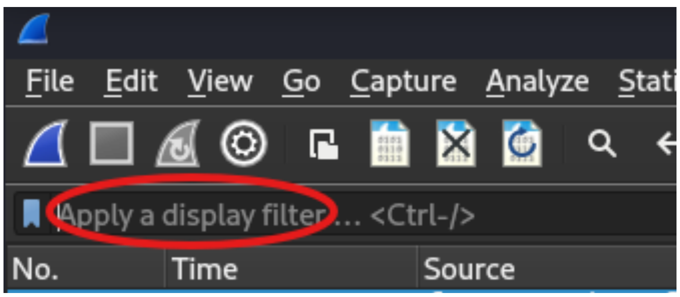
Check http://www.ascii-code.com/, that is, the ASCII table:

Provide the binary representation of the character A (letter), the character 3 (number) and the no-displayable character \r (carriage return).

Using the ASCII table, find out the Hexadecimal representations of the characters G, E and T (that is the word GET).

Exercise 3: Wireshark Display/Filtering

Continuing with the captured packets on Wireshark, type http in the tool bar text box and click Apply. The background of the text box turns green when the typed text in the correct syntax (that is, expected by Wireshark). Wireshark should display Frames with http protocol (as the top protocol).



Type TCP in the tool bar text box and click Apply.

☐ Q8 - Why are there some frames labelled as HTTP under the protocol heading after applying the tcp filter?

If you need to exclude the HTTP from the TCP frames, you can type "tcp and not http" (which means filter for tcp and exclude http) in the filter text box. Try it out. **Provide your results.** For more information regarding Wireshark filters check the following link: WireShark Docs: Building Display Filter Expressions.

If you need to display packets sent (sourced from) by a specific IP address, you should enter **ip.src== "ip address"** (replace the word "ip address" with the specific IP address you are searching for). **Provide an example**. To find out the entire set of Display Filter syntax, you may click the Expression button next to the filter text box. For example, scroll to the IP entry and expand by clicking on the + sign or the arrow on the left. Wireshark should display all possible attributes of the object IP. Find 'ip.ttl' and find out what it means. Type 'ip.ttl' then '==' and type '64' in the value field so that the equation would look like the following (ip.ttl == 64) and Click OK.

Q9 - Describe what happens (what you have witnessed). Q10 – Can I filter for a specific source IP address and specific destination IP address? Provide screenshots.